



F500 Document
Capstone Design
Implementation
Driving SME Growth with AI-Based Seller
Support Solutions for Selly

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Submitted for
Capstone Design Project
to Faculty of Computer Science
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STATEMENT OF ORIGINALITY

In my capacity as an active student at President University and as the author of the Capstone Design Project stated below:

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Faculty : Computer Science

I hereby declare that my Capstone Design Project entitled “**Driving SME Growth with AI Based Seller Support Solutions for Selly**” is to the best of my knowledge and belief, an original piece of work based on sound academic principles. If there is any plagiarism detected in this final project, I am willing to be personally responsible for the consequences of these acts of plagiarism and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, 27 June 2025

Signer 1	Signer 2
	
Joseph Tedja Nugraha Wibawa – 001202200121	Jaya Iskandar MF– 001202200042

Driving SME Growth with AI-Based Seller Support Solutions for Selly

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PART 5 TESTING (F500)

Consists of:

- A. Restate the problem as stated in the F-100 document (PART A in F100)**
- B. Restate the specifications stated in the F-200 document (PART B in F200).**
- C. Functional testing**
 - 1. Testing results of every function in the specification (based on PART C in F100, PART C NO.4 in F200, PART D in F200)**
 - 2. Present qualitative testing (example: questionnaire)**
 - 3. Detail the test procedures that are carried out according to the design.**
 - 4. Procedures for the demo are created and verified.**
- D. Testing other specifications:**
 - 1. Non-functional specifications such as size, weight, etc., that are included in the document (PART C NO. 5 F200)**
 - 2. A photo/recording of the test is shown in the document**

A. RESTATE THE PROBLEM AS STATED IN THE F-100 DOCUMENT (PART A IN F100)

1. Background of the Problem

For years, SME (Small and Medium Enterprises) business owners have been increasing their usage of online channels like WhatsApp, Instagram, and Facebook and shifting from the old traditional selling method at physical retail stores. However, managing customer interactions has several challenges, e.g., responding to repetitive inquiries, processing transactions, and consistent communication. For instance, sellers often spend hours manually answering the same customer queries, tracking orders, or reminding customers about pending payments.

With such a scenario, the app Selly — a part of GoTo Financial — was launched in 2020 to serve as the “in-built smart keyboard” for automating all of these business processes, starting from an auto-template text message, courier price checker, automated invoice, till automated reminder. Selly has offered impressive integration between required platforms such as payment gateways and couriers while managing in-store information within the app.

While it managed to address basic automation needs, business-wise, it failed to keep up with the evolving changes of the market as most owners have shifted to big marketplaces, e.g., Tokopedia and Shopee, which have all-in-one integrated features needed for these owners, including its auto text message, thereby “diminishing” the value of Selly itself. Existing tools like Selly are outdated, failing to integrate the necessary automation and intelligence required to meet the needs of modern SME sellers.

2. Problem Statement

Selly has become outdated and lacks the intelligent, AI-driven tools needed to stay relevant in a competitive market. Its limited automation and static features fail to address the dynamic demands of modern sellers.

3. Objective

There are three (3) main objectives of this Capstone project:

- Increase User Adoption (Brand Awareness) of Selly
Enhance Selly to make it more appealing and relevant to users by making it more “powerful”, encouraging more SME owners to use the platform.
- Add Compelling Features to Selly
Integrate with new AI features to boost Selly’s value proposition for the users in assisting their business operations.
- Support SME Growth
Provide tools that help SMEs save time, improve customer interactions, and focus on growing their businesses, making Selly a key part of their success.

4. The problem from a Customer Perspective

Sellers feel that Selly is not as helpful as it could be because they could have managed everything within large marketplace apps, e.g., Tokopedia and Shopee, which provide all-in-one solutions for communication and transaction management. Additionally, many sellers find it inconvenient or unfamiliar to handle these tasks through a smart keyboard instead of a dedicated app, leading to a preference for other tools.

B. RESTATE THE SPECIFICATIONS STATED IN THE F-200 DOCUMENT.

1. Main Functionality

The primary functionalities of the system are:

- Business-related knowledge base input and vectorization: Uses an input and vectorization mechanism to allow sellers to generate a vectorized representation of their business-related knowledge base
- General command input: Uses an input and appendix mechanism to allow sellers to attach a command directed to the agent of the interaction manner with the customer.
- Similar top-n sections retrieval: The system retrieves relevant sections of the knowledge base with a similarity measure (e.g. cosine similarity).
- LLM-powered auto-text responses: Uses a knowledge base to generate dynamic and contextually relevant replies for customer queries.
- Personalized marketing campaign: Generates targeted marketing messages using AI-driven customer segmentation.

2. User Characteristics

The system caters to 2 primary user types: SME Sellers and Customers.

- SME Sellers: These users are responsible for managing customer interactions and sales through the platform. They have full access to the system, allowing them to set up AI-powered auto-text responses, monitor customer engagement, and customize marketing campaigns. Their education level is generally high school or above, with basic digital literacy skills. Most have moderate experience in online selling, and they undergo basic platform training to familiarize themselves with AI-powered tools.
- Customers: These are the end-users who interact with the seller through the chat interface. They do not have direct access to the system but benefit from improved response times and personalized interactions. Since customers do not require any

operational control over the system, their education level, skill level, experience, and training are not applicable (N/A).

Users	Responsibility	Access rights	Education Levels	Skill Levels	Experience	Type of Training
SME Seller	Manage customer interaction & sales	Full access	High school / above	Basic digital literacy	Moderate	Basic platform training
Customer	Interacts with the seller via chat	No access	N/A	N/A	N/A	N/A

3. Constraints

The following are the constraints of the system:

- **Technical Constraints:** The user-end system must work efficiently on low-end devices with limited processing power.
- **Privacy Constraints:** Compliance with Indonesian data protection regulations to ensure user data security and privacy.
- **Adoption Constraints:** The system must be intuitive and easy to use, preventing a steep learning curve for SME sellers.
- **Connectivity Constraints:** Requires an internet connection for AI-powered automation and cloud storage.

4. Product Development Environment

- a. **Hardware:** Cloud servers, mobile devices (Android, iOS), developer workstations.
- b. **Software:** Python, TensorFlow, Android Studio or XCode, LangChain, Node.js, PostgreSQL, Pinecone or PgVector.
- c. **Connection:** Internet connectivity is required for real-time AI model execution and message automation.

5. Product Operational Environment

- a. Hardware: Smartphones, cloud storage servers, API gateways.
- b. Software: The frontend application should be mobile-based, while the backend application is hosted on a cloud server.
- c. Connection: Requires stable internet for real-time processing and AI automation.

C. FUNCTIONAL TESTING

1. Testing results of every function in the specification

a. Part C in F100

i. User Authentication

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Register	The seller will open the application, navigate to the register page, fill in the registration form, and click on the register button	(10.30.00) The Seller opens the application (10.30.03) Seller clicks the <i>Register Here</i> text on the login page (10.30.04) A form will appear (10.30.15) Seller fills in the data (email and password) (10.30.16) Seller clicks on the register button	Data from the registration form will be sent to the backend, verification will be done, and if there are no issues, the user will be navigated to the homepage	The application will display a homepage and a success message at the bottom of the screen (as a toast)	01/06/2025	Success
Positive Testing	Log out	The seller will open the application and register/log in/	(10.30.20) Seller opens the application (10.30.22)	Persisted data on the device will be deleted,	The application will display the login	01/06/2025	Success

		be navigated to the home screen through a persistent session	Seller presses the logout button on the top right side of the page	and the seller will need to re-login to use the application	page.		
Negative Testing	Session Refresh	The seller will open the application after logging out and will be positioned at the login page	(10.30.24) Seller reopens the application after logging out	As there are no persisted data left, the seller should stay on the login page	The application will display the login page.	01/06/2025	Success
Negative Testing	Register	The seller will open the application, navigate to the register page, fill in the registration form with invalid or incomplete data, and click on the register button.	(10.30.25) The Seller opens the application (10.30.35) Seller clicks the <i>Register here</i> text on the login page (10.30.36) The registration form appears (10.30.40) Seller attempts to submit the form with one or more invalid inputs (either email	The seller will receive an error message due to the invalid information	The application will show an error message due to the invalid information, and keep the seller on the register page	01/06/2025	Success

			or password)				
Negative Testing	Log in	Log in with invalid or incomplete information	(10.30.45) The Seller opens the application (10.30.46) The login form appears (10.30.50) Seller attempts to submit the form with one or more invalid inputs (either email or password)	The seller will receive an error message due to the invalid information	The application will show an error message due to the invalid information, and keep the seller on the login page	01/06/2025	Success
Positive Testing	Log in	Log in with valid information	(10.30.55) The Seller opens the application (10.31.00) The login form will appear (10.31.05) Seller fills in valid data (email and password)	Data from the login form will be sent to the backend, verification will be done, and if there are no issues, the user will be navigated to the homepage	The application will display a homepage and a success message at the bottom of the screen (as a toast)	01/06/2025	Success
Positive	Session	Reopen the	(10.31.07)	As there is	The	01/06/2025	Success

Testing	Refresh	application that was closed without logging out	Seller reopens the application after closing it without logging out	persistent data left, the seller should be navigated to the home page	application will display the home page.		
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ii. Business Knowledge Base Input

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Add default knowledge base data	Generate the default knowledge base information	(10.31.10) The Seller opens the application (10.31.13) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.31.14) Seller clicks on the Knowledge base tab on the top tab layout (10.31.15) Seller clicks on the “Let’s make one now” button in	Data from the default knowledge base will be used to create a new knowledge base, and the user will be presented with the default knowledge base that was just created. The first knowledge base that is	The application will display the knowledge base list page with the newly created default knowledge base and present a success message at the bottom of the screen (as a toast). The first	01/06/2025	Success

			the middle of the screen	owned by a user is automatically activated	knowledge base that is owned by a user is automatically activated		
Positive Testing	Add knowledge e-base data	Submit valid knowledge base information	(10.31.17) The Seller opens the application (10.31.19) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.31.21) Seller clicks on the Knowledge base tab on the top tab layout (10.31.23) Seller clicks on the '+' icon on the bottom right side of the screen (10.31.33) Seller fills in the create knowledge base form	Data from the knowledge base creation form will be sent to the backend, stored in the database, and the user will be navigated to the knowledge base tab of the RAG menu. If the knowledge base that was created is the first knowledge base owned by the user, it is	The application will display the knowledge base list page with the newly created item present and a success message at the bottom of the screen (as a toast)	01/06/2025	Success

			(10.31.34) Seller clicks on the submit button	automaticall y activated			
Negative Testing	Add knowledg e-base data	Submit invalid or incomplete knowledge base information	(10.31.35) The Seller opens the application (10.31.37) Seller clicks the RAG menu on the bottom navigation bar (10.31.39) Seller clicks on the Knowledge base tab on the top tab layout (10.31.41) Seller clicks on the '+' icon on the bottom right side of the screen (10.31.42) Seller leaves either all fields 'Knowledge base name', or the 'Knowledge base content'	Client-side validation should prevent submission.	An error message will be displayed next to the 'Knowledge base name' field (e.g., "Knowledg e base name can't be empty" or "Knowledg e base content can't be empty"). User remains on the create knowledge base form.	01/06/2025	Success

			field empty (10.31.45) Seller clicks on the submit button				
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iii. Business Knowledge Base Management

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	View knowledge base data	View the list of all knowledge bases	(10.31.50) The Seller opens the application (10.31.52) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.31.54) Seller clicks on the Knowledge base tab on the top tab layout	Data from all knowledge bases owned by the user is displayed	The application will display the knowledge base list page with all previously created knowledge base items	01/06/2025	Success
Positive Testing	View knowledge base detail	View the detailed information of a knowledge base	(10.31.56) The Seller opens the application (10.31.58) Seller clicks	Detail of a knowledge base owned by the user is displayed	The application will display the knowledge base name	01/06/2025	Success

			the <i>RAG</i> menu on the bottom navigation bar (10.32.00) Seller clicks on the Knowledge base tab on the top tab layout (10.32.02) The seller clicks on any existing knowledge base item		as well as the list of chunks associated with the clicked knowledge base item		
Positive Testing	Activate the knowledge base item	Activate a knowledge base	(10.32.04) The Seller opens the application (10.32.06) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.32.08) Seller clicks on the Knowledge base tab on the top tab layout (10.32.10) Seller clicks	The selected knowledge base item is activated while the previously active item is deactivated	The application will display the activated knowledge base item at the very top of the list of knowledge bases on the knowledge base list page and the previously activated	01/06/2025	Success

			the three vertical dots button on any existing knowledge base item (10.32.12) The seller is shown a small menu near the item (10.32.14) Seller clicks on the activate option on the small menu		item is deactivated		
Positive Testing	Delete knowledge e-base	Delete knowledge base item	a (10.32.16) The Seller opens the application (10.32.18) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.32.20) Seller clicks on the Knowledge base tab on the top tab layout	Knowledge base item is deleted	The application will display the list of knowledge base items that already have the deleted item removed from the list	01/06/2025	Success

			(10.32.22) Seller clicks the three vertical dots button on any existing knowledge base item (10.32.24) The seller is shown a small menu near the item (10.32.26) Seller clicks on the delete option on the small menu (10.32.28) The seller is shown a deletion confirmation dialog (10.32.30) Seller clicks on the delete option on the dialog				
Negative Testing	Delete knowledge base	Cancel the deletion of a knowledge	(10.32.32) The Seller opens the	Knowledge base item deletion is	The application will display	01/06/2025	Success

		base item on the deletion confirmation dialogue	application (10.32.34) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.32.36) Seller clicks on the Knowledge base tab on the top tab layout (10.32.38) Seller clicks the three vertical dots button on any existing knowledge base item (10.32.40) The seller is shown a small menu near the item (10.32.42) Seller clicks on the delete option on the small menu (10.32.42) The seller is shown a	aborted	the previous list of knowledge base items (the item which deletion is cancelled is still present on the list)		
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			deletion confirmation dialog (10.32.44) Seller clicks on the cancel option on the dialog				
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iv. Knowledge Base Chunks Management

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	View chunk detail data	View the details of the knowledge base's chunk	(10.32.46) The Seller opens the application (10.32.48) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.32.50) Seller clicks on the Knowledge base tab on the top tab layout (10.32.52) The seller clicks on any existing	The content of the chunk is displayed on the chunk detail page	The application will display the chunk detail on the chunk detail page	01/06/2025	Success

			knowledge base item (10.32.54) The seller clicks on any existing chunk item				
Positive Testing	Edit chunk data	Edit the contents of the knowledge base's chunk	(10.32.56) The Seller opens the application (10.32.58) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.33.00) Seller clicks on the Knowledge base tab on the top tab layout (10.33.02) The seller clicks on any existing knowledge base item (10.33.04) The seller clicks on any existing chunk	Updated content of the chunk is displayed on the chunk detail page	The application will display the updated chunk detail on the chunk detail page	01/06/2025	Success

			item (10.33.06) The seller clicks on the edit icon button on the top right side of the screen (10.33.08) The seller modifies the chunk information displayed in the field (10.33.09) The seller clicks on the submit button				
Negative Testing	Edit chunk data	Empty the contents of an existing knowledge base's chunk.	(10.33.10) The Seller opens the application (10.33.12) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.33.14) Seller clicks on the Knowledge	An error message, “Chunk content cannot be empty,” will be displayed on the chunk content field	The application will display the error message and keep the user on the chunk edit page	01/06/2025	Success

			base tab on the top tab layout (10.33.16) The seller clicks on any existing knowledge base item (10.33.18) The seller clicks on any existing chunk item (10.33.20) The seller clicks on the edit icon button on the top right side of the screen (10.33.22) The seller empties the chunk of information displayed in the field (10.33.23) The seller clicks on the submit button				
Positive	Delete	Delete	an (10.33.24)	The chunk	The	01/06/2025	Success

Testing	chunk data	existing knowledge base's chunk.	The Seller opens the application (10.33.26) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.33.28) Seller clicks on the Knowledge base tab on the top tab layout (10.33.30) The seller clicks on any existing knowledge base item (10.33.32) The seller clicks on the delete button on any of the existing chunk items	item is deleted	application will display the chunk list page with the deleted item excluded, and a message will appear at the bottom of the screen notifying that the delete is successful.		
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v. Knowledge Base Chunks Input

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
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Positive Testing	Create a new chunk	Create a new chunk on an existing knowledge base on the knowledge base detail page/chunk list page	(10.33.34) The Seller opens the application (10.33.36) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.33.38) Seller clicks on the Knowledge base tab on the top tab layout (10.33.40) The seller clicks on the + icon on the bottom right side of the screen (10.33.50) The seller inputs the new chunk content into the chunk content field (10.33.52) The seller clicks on the submit button	The new chunk is displayed together with all previously existing chunks on the chunk list page	The application will display the newly created chunk together with all previously existing chunks on the chunk list page	01/06/2025	Success
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Negative Testing	Create a new chunk	Create a new chunk on an existing knowledge base on the knowledge base detail page/chunk list page, but with empty content	(10.33.54) The Seller opens the application (10.33.56) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.33.58) Seller clicks on the Knowledge base tab on the top tab layout (10.34.00) The seller clicks on the + icon on the bottom right side of the screen (10.34.02) The seller clicks on the submit button	An error message, “Chunk content cannot be empty,” will be displayed on the chunk content field	The application will display the error message and keep the user on the chunk creation page	01/06/2025	Success
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vi. Agent Input

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive	Add	Generate the	(10.34.04)	Data from	The	01/06/2025	Success

Testing	default agent data	default agent information	The Seller opens the application (10.34.06) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.34.08) Seller clicks on the Agent tab on the top tab layout (10.34.10) Seller clicks on the “Let’s make one now” button in the middle of the screen	the default agent will be used to create a new agent, and the user will be presented with the default agent that was just created.	application will display the agent list page with the newly created default agent and present a success message at the bottom of the screen (as a toast)		
Positive Testing	Add agent data	Submit valid agent information	(10.34.12) The Seller opens the application (10.34.14) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.34.16) Seller clicks on the Agent	Data from the agent creation form will be sent to the backend, stored in the database, and the user will be navigated to the agent	The application will display the agent list page with the newly created item present and a success message at the bottom	01/06/2025	Success

			tab on the top tab layout (10.34.18) Seller clicks on the '+' icon on the bottom right side of the screen (10.34.20) Seller fills in the create agent form (10.34.22) Seller clicks on the submit button	tab of the RAG menu	of the screen (as a toast)		
Negative Testing	Add agent data	Submit invalid or incomplete agent information	(10.34.24) The Seller opens the application (10.34.26) Seller clicks the RAG menu on the bottom navigation bar (10.34.28) Seller clicks on the Agent tab on the top tab layout (10.34.30) Seller clicks on the '+' icon	Client-side validation should prevent submission.	An error message will be displayed next to the 'Agent name' field (e.g., "Agent name can't be empty" or "Agent content can't be empty"). User	01/06/2025	Success

			on the bottom right side of the screen (10.34.32) Seller leaves either all fields 'Agent name', or the 'Agent content' field empty (10.34.34) Seller clicks on the submit button		remains on the create agent form.		
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vii. Agent Management

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	View agent data	View the list of all agents	(10.34.36) The Seller opens the application (10.34.38) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.34.40) Seller clicks on the Agent tab on the top	Data from all agents owned by the user is displayed	The application will display the agent list page with all previously created agent items	01/06/2025	Success

			tab layout				
Positive Testing	View agent detail	View the detailed information of an agent	(10.34.42) The Seller opens the application (10.34.44) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.34.46) Seller clicks on the Agent tab on the top tab layout (10.34.48) The seller clicks on any existing agent item	Detail of an agent owned by the user is displayed	The application will display the agent name as well as the agent command, and the temperature of the clicked agent item	01/06/2025	Success
Positive Testing	Activate the agent item	Activate an agent	(10.34.50) The Seller opens the application (10.34.52) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.34.54) Seller clicks on the Agent	The selected agent item is activated while the previously active item is deactivated	The application will display the activated agent item at the very top of the list of agents on the agent list page,	01/06/2025	Success

			tab on the top tab layout (10.34.56) Seller clicks the three vertical dots button on any existing agent item (10.34.58) The seller is shown a small menu near the item (10.35.00) Seller clicks on the activate option on the small menu		and the previously activated item is deactivated		
Positive Testing	Delete agent	Delete an agent item	(10.35.02) The Seller opens the application (10.35.04) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.35.06) Seller clicks on the Agent tab on the top	The agent item is deleted	The application will display the list of agent items that already have the deleted item removed from the list	01/06/2025	Success

			tab layout (10.35.08) Seller clicks the three vertical dots button on any existing agent item (10.35.10) The seller is shown a small menu near the item (10.35.12) Seller clicks on the delete option on the small menu (10.35.14) The seller is shown a deletion confirmation dialog (10.35.26) Seller clicks on the delete option on the dialog				
Negative Testing	Delete agent	Cancel the deletion of an agent item on	(10.35.28) The Seller opens the	Agent item deletion is aborted	The application will display	01/06/2025	Success

		the deletion confirmation dialogue	application (10.35.30) Seller clicks the <i>RAG</i> menu on the bottom navigation bar (10.35.32) Seller clicks on the Agent tab on the top tab layout (10.35.34) Seller clicks the three vertical dots button on any existing agent item (10.35.36) The seller is shown a small menu near the item (10.35.38) Seller clicks on the delete option on the small menu (10.35.40) The seller is shown a deletion confirmation		the previous list of agent items (the item which deletion is cancelled is still present on the list)		
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			dialog (10.35.42) Seller clicks on the cancel option on the dialog				
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viii. Keyboard Interface Usage

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Negative Testing	Install Keyboard Interface	Abort the installation of the Selly Keyboard interface on the device	(10.34.44) The Seller opens the application (10.34.46) Seller clicks the keyboard icon on the top right side of the home page (10.34.48) A change system keyboard dialogue will appear on the page (10.34.50) The seller clicks on the	The installation of the Selly keyboard interface is aborted	The seller will be directed back to the home page. On the next text input focus, the previous keyboard would still appear instead of the Selly Keyboard Interface	01/06/2025	Success

			cancel button on the dialogue				
Negative Testing	Install Keyboard Interface	Install the Selly Keyboard interface, but activate another keyboard interface	(10.35.52) The Seller opens the application (10.35.54) Seller clicks the keyboard icon on the top right side of the home page (10.35.56) A change system keyboard dialogue will appear on the page (10.35.58) The seller clicks on the install keyboard button on the dialogue (10.36.00) A native, choose input method dialog will appear on	The selected keyboard interface would appear when an input is made on the device.	On the text input focus, the selected keyboard interface would appear	01/06/2025	Success

			the page (10.36.02) The seller clicks on any option other than Selly				
Positive Testing	Install Keyboard Interface	Install the Selly Keyboard interface on the device	(10.36.04) The Seller opens the application (10.36.06) Seller clicks the keyboard icon on the top right side of the home page (10.36.08) A change system keyboard dialogue will appear on the page (10.36.10) The seller clicks on the install keyboard button on the dialogue (10.36.12) A native,	The Selly keyboard interface would appear when an input is made on the device, replacing the previous keyboard interface of the device	On the text input focus, the Selly keyboard interface would appear instead of the default keyboard interface of the device	01/06/2025	Success

			choose input method dialog will appear on the page (10.36.14) The seller clicks on the Selly app option on the dialog				
Positive Testing	Query to the RAG system on the Selly keyboard interface	Query to the RAG system on the Selly keyboard interface	(10.36.04) The Seller opens the social media application with a text input (10.36.06) The seller selects the customer to converse with and enters the chat interface (10.36.08) Seller presses on the text box (10.36.10) The seller copies the customer's question to the	The seller should be able to send the adjusted and contextually correct RAG result to the customer.	An agent command compliant and contextually correct to the knowledge base answer that can be modified and sent to the customer by the seller	01/06/2025	Success

			text box (10.36.12) The seller presses the “ask RAG” button above all the keyboard key items (10.36.30) The seller waits for a few moments while the system does the similarity measurement and answer generation. The wait time depends on the question’s complexity and the possible answer length (10.36.32) The “ask RAG” button turns into a copy button (10.36.34) The seller				
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			clicks on the copy button (10.36.36) The seller pastes the RAG system result into the text box. (10.36.38) If needed, the seller can modify the RAG system result in the text box. (10.36.40) The seller sends the finalized result to the customer				
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ix. Invoice Analysis Mechanism

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Create new invoice	Submit valid invoice information	(14.30.00) The Seller opens the application (14.30.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.30.04) Seller clicks on the Invoices tab on the top tab layout (14.30.06) Seller clicks on the '+' icon on the bottom right side of the screen (14.30.36) Seller fills in the create invoice form (14.30.40) Seller clicks on the submit	Data from the invoice creation form will be sent to the backend, stored in the database, and the user will be navigated to the invoice tab of the Marketing menu	The application will redirect user to the invoices list screen, indicating that the new invoice has been successfully created & can be found at the bottom of the list.	01/06/2025	Success

			button				
Negative Testing	Create new invoice	Submit invalid invoice information (non-standard phone number; empty products list)	(14.31.00) The Seller opens the application (14.31.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.31.04) Seller clicks on the Invoices tab on the top tab layout (14.31.06) Seller clicks on the '+' icon on the bottom right side of the screen (14.31.36) Seller fills in the create invoice form, filling non-standard phone number / empty products list (14.31.40)	Client-side validation should prevent submission.	An error message will be displayed e.g., Product must be >1" or "phone number must begin with 08xx and 10-12 digits"). User remains on the create invoice form.	01/06/2025	Success

			Seller clicks on the submit button				
Positive Testing	View invoice data	View the list of all invoices	(14.32.00) The Seller opens the application (14.32.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.32.04) Seller clicks on the Invoices tab on the top tab layout	Data from all invoices owned by the user is displayed	The application will display the invoice list page with all previously created invoice items	01/06/2025	Success
Positive Testing	View invoice detail	View the detailed information of an invoice	(14.33.00) The Seller opens the application (14.33.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.33.04) Seller clicks on the Invoices tab on the top	Detail of an invoice owned by the user is displayed	The application will display the customer details with the products list	01/06/2025	Success

			tab layout (14.33.10) The seller clicks on any existing invoice item				
Positive Testing	Update invoice detail	Edit the contents of an invoice	(14.34.00) The Seller opens the application (14.34.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.34.04) Seller clicks on the Invoices tab on the top tab layout (14.34.06) The seller clicks on any existing invoice (14.34.30) The seller writes over the current populated data, e.g., updating	Updated content of the invoice is displayed on the invoice detail page	The application will display the updated invoice detail on the invoice detail screen	01/06/2025	Success

			phone number, or products & prices. (14.34.32) The seller clicks on the save button				
Positive Testing	Delete invoice	Delete invoice	an (14.35.00) The Seller opens the application (14.35.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.35.04) Seller clicks on the Invoices tab on the top tab layout (14.35.06) Seller clicks on any existing invoice (14.35.10) The seller clicks on the delete button of the invoice detail form	Invoice is deleted	The application will display the list of invoices that already have the deleted item removed from the list	01/06/2025	Success

Positive Testing	Conduct Cluster-ing	Clusterize customers' invoices	(14.40.00) The Seller opens the application (14.40.02) Seller clicks the <i>Marketing</i> menu on the bottom navigation bar (14.40.04) Seller clicks on the Clustering tab on the top tab layout (14.40.06) The pop-up of start clustering shows up on the centre of screen (14.40.08) Seller clicks on 'proceed' button	Clustering will be processed by the backend server, results sent to the database, and the user will be navigated to the cluster results screen, seeing the table and list.	The application will redirect user to the cluster results screen, displaying in the form of table and list.	01/06/2025	Success
Negative Testing	Conduct Cluster-ing	Clusterize customers' invoices	(14.41.00) The Seller opens the application (14.41.02) Seller clicks	Client-side validation should prevent clustering.	An error message will be displayed e.g., "no. of invoices	01/06/2025	Success

			the <i>Marketing</i> menu on the bottom navigation bar (14.41.04) Seller clicks on the Clustering tab on the top tab layout (14.41.06) Message restricting user to proceed due to number of invoices <3 (bare minimum is 3; optimum >30)		must be >3". User remains on the screen; forcing the user to create more invoices.		
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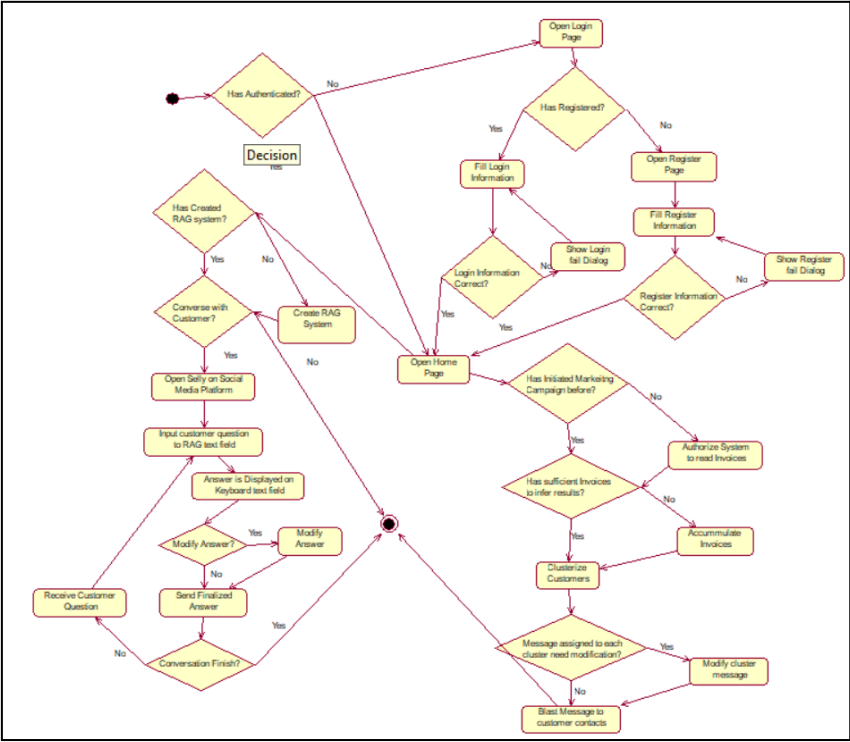
x. Automated Marketing Campaign

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Generate AI-suggested offer message	Generate the message	(14.45.00) The seller has finished reaching the cluster result	The AI message would appear onto the fields	The application instantly populates the text	01/06/2025	Success

			page (14.45.04) The seller clicks on ‘save’ and is brought to Campaign page (14.45.06) The seller sees the list of customer-message text field (14.45.10) The seller clicks on the generate AI offer button		fields with AI-generated message		
Positive Testing	Editing offer message	Modifying the message	(14.45.20) The seller modifies manually the text fields, be it either when it was blank priorly or upon populated by the AI message	It would overwrite and update the current field	The application instantly overwrites with the user’s input	01/06/2025	Success
Positive Testing	Sending offer message	WhatsApp the message	(14.45.30) The seller chooses the	User would be brought to	The application redirects	01/06/2025	Success

			target recipient (14.45.36) The seller clicks on “Send via WhatsApp button	WhatsApp to proceed with sending it	user to WhatsApp with the copied content of its message onto the recipient’s conversa- tion.		
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b. Part C No. 4 in F200



The scenarios above have already been tested in the prior section.

c. Part D in F200

Response Time: The program should generate a response within 40 seconds for AI-generated replies

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Query to the RAG System	Query to the RAG system on the Selly keyboard interface	(10.57.00) The Seller opens the social media application that has a text input (10.57.02) The seller selects the customer to converse with and enters the chat interface (10.57.08) Seller presses on the text box (10.57.10) The seller copies the customer's question to the text box (10.57.12) The seller	Results are given in less than 40 seconds	The application will allow the user to copy the result	02/06/2025	Success (Only about 5 seconds are taken)

			presses the “ask RAG” button above all the keyboard key items (10.57.18) The seller waits for a few moments while the system does the similarity measurement and answer generation. The wait time depends on the question’s complexity and the possible answer length (10.57.32) The “ask RAG” button turns into a copy button (10.57.34) The seller clicks on the copy button (10.57.36)				
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			The seller pastes the RAG system result into the text box. (10.57.38) If needed, the seller can modify the RAG system result in the text box. (10.57.40) The seller sends the finalized result to the customer				
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The recording that proves the duration taken can be accessed at the google drive link

<https://drive.google.com/file/d/1DNCvid40-5ArM9DEdQxYj71Bshh3CaZM/view?usp=sharing>

Accuracy of AI Responses: AI-generated responses should achieve at least 85% accuracy, based on manual review alignment of a knowledge base that has more than 10 chunks.

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive	Query to	Query to the	(09.30.00)	Data from	The	02/06/2025	Success

Testing	the RAG System	RAG system on the Selly keyboard interface	The Seller opens the social media application that has a text input (09.30.02) The seller selects the customer to converse with and enters the chat interface (09.36.08) Seller presses on the text box (09.36.10) The seller copies the customer's question to the text box (09.36.12) The seller presses the "ask RAG" button above all the keyboard key items (09.36.30) The seller waits for a few	all agents owned by the user is displayed	application will display the agent list page with all previously created agent items		
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			moments while the system does the similarity measurement and answer generation. The wait time depends on the question's complexity and the possible answer length (09.36.32) The “ask RAG” button turns into a copy button (09.36.34) The seller clicks on the copy button (09.36.36) The seller pastes the RAG system result into the text box. (09.36.38) If needed, the seller can modify the				
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			RAG system result in the text box. (09.36.40) The seller sends the finalized result to the customer				
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Agent Command:

“You're a store assistant whose task is to answer customer questions regarding a store. Do not go out of topic and follow the given knowledge.

Make sure your answer is within 150 words.”

The Daily Grind knowledge base (12/12 chunks and 1 out of context query tested)

No	Targeted Chunk	Question	RAG Response	Status
1.	# Store Overview Our store, "The Daily Grind," is a community-focused coffee shop and artisanal bakery located at 123 Main Street, Anytown, USA. We pride ourselves on serving high-quality, ethically sourced coffee, freshly baked goods made from scratch daily, and providing a welcoming atmosphere for our customers. Our mission is	What is the mission of The Daily Grind and what kind of atmosphere does it aim to provide?	The mission of The Daily Grind is to be the neighborhood's favorite spot for a morning pick-me-up, a quick lunch, or a relaxed afternoon treat. We aim to provide a welcoming atmosphere for our customers, serving high-quality, ethically sourced coffee and freshly baked goods made	Success

	to be the neighborhood's favorite spot for a morning pick-me-up, a quick lunch, or a relaxed afternoon treat. We believe in fostering a sense of community and supporting local suppliers whenever possible.		from scratch daily. We also strive to foster a sense of community and support local suppliers whenever possible. Our store features comfortable seating, free Wi-Fi, and a curated playlist to enhance the relaxed atmosphere.	
2.	# Business Hours The Daily Grind operates seven days a week to best serve our community. * Monday - Friday: 6:00 AM - 7:00 PM * Saturday: 7:00 AM - 6:00 PM * Sunday: 7:00 AM - 5:00 PM Please note that holiday hours may vary and will be posted prominently in-store and on our social media channels. We recommend checking our website or social media for the most up-to-date	Can you tell me the business hours of the daily grind?	Certainly! The Daily Grind operates as follows: - Monday to Friday: 6:00 AM to 7:00 PM - Saturday: 7:00 AM to 6:00 PM - Sunday: 7:00 AM to 5:00 PM Please note that holiday hours may vary, so it's a good idea to check our website or social media for the most up-to-date information.	Success

	information on public holidays.			
3.	# Contact Information For any inquiries, feedback, or to place an order, please use the following contact information: * Phone: (555) 123-4567 * Email: info@thedailygrind.com * Website: www.thedailygrind.com * Physical Address: 123 Main Street, Anytown, USA Our team is dedicated to responding to all inquiries within 24-48 business hours. For urgent matters, please call us directly during business hours.	How can I contact the Daily Grind?	You can contact The Daily Grind through the following methods: - Phone: (555) 123-4567 - Email: info@thedailygrind.com - Website: www.thedailygrind.com - Physical Address: 123 Main Street, Anytown, USA Our team aims to respond to all inquiries within 24-48 business hours. For urgent matters, please call us directly during business hours.	Success
4.	# Product Offerings: Coffee Our coffee selection at The Daily Grind is curated to offer a diverse range of flavors and brewing methods. We source	What are the coffee types that I can order at the Daily Grind?	At The Daily Grind, you can choose from a variety of coffee types, including espresso-based drinks like lattes,	Success

	our beans from sustainable and fair-trade farms around the world, ensuring both quality and ethical practices. Our standard coffee menu includes: * Espresso-based drinks: Latte, Cappuccino, Americano, Macchiato, Flat White, Mocha * Drip Coffee: Rotating daily brew, cold brew * Pour Over: Single-origin options available, updated weekly * Iced Coffee: Our signature iced coffee and various iced espresso beverages We also offer a selection of dairy alternatives (oat, almond, soy) and flavor syrups.		cappuccinos, americanos, macchiatos, flat whites, and mochas. We also offer drip coffee with a daily rotating brew and cold brew options. For a more personalized experience, try our pour-over coffee with single-origin beans that change weekly. Additionally, we serve iced coffee and various iced espresso beverages. We have dairy alternatives and flavor syrups available to customize your drink.	
5.	# Product Offerings: Bakery At The Daily Grind, our bakery items are baked fresh daily, from scratch, using high-quality ingredients. Our bakers arrive early each	What fresh bakery items are typically available at The Daily Grind, and are there any seasonal or holiday	At The Daily Grind, our bakery items are baked fresh daily using high-quality ingredients. Our typical selection includes:	Success

	morning to prepare a delicious array of pastries, breads, and desserts. Our daily bakery selection typically includes: * Croissants: Butter, chocolate, almond * Muffins: Blueberry, chocolate chip, banana nut * Scones: Berry, cheese * Cookies: Chocolate chip, oatmeal raisin, snickerdoodle * Cakes & Slices: Daily rotating specials * Artisanal Bread: Sourdough, whole wheat (available certain days) We also feature seasonal and holiday-themed baked goods. Inquire with our staff about today's fresh selections!	specials?	- Croissants: Butter, chocolate, and almond - Muffins: Blueberry, chocolate chip, and banana nut - Scones: Berry and cheese - Cookies: Chocolate chip, oatmeal raisin, and snickerdoodle - Cakes & Slices: Daily rotating specials - Artisanal Bread: Sourdough and whole wheat (available on certain days) We also offer seasonal and holiday-themed baked goods. For the latest fresh selections, feel free to ask our staff!	
6.	# Loyalty Program The Daily Grind values our loyal customers, and we've created a rewarding loyalty program to show our appreciation. Our "Grind Rewards" program is free to	How does "Grind Rewards" loyalty program work, and what benefits do members receive?	The "Grind Rewards" loyalty program at The Daily Grind is free to join and offers several benefits. Members earn 1 point for every \$1 spent, and 100 points can be	Success

	join and offers exclusive benefits. How it works: * Earn 1 point for every \$1 spent. * Redeem 100 points for a free standard coffee. * Receive a special treat on your birthday. * Get exclusive offers and early access to new products. Sign up at the counter during your next visit or through our website. Your points never expire!		redeemed for a free standard coffee. Additionally, members receive a special treat on their birthday and get exclusive offers and early access to new products. You can sign up at the counter during your next visit or through our website, and your points never expire.	
7.	# Catering Services Looking to add a delicious touch to your next event? The Daily Grind offers catering services for both coffee and baked goods. Whether it's a corporate meeting, a family gathering, or a special celebration, we can provide fresh, high-quality items to impress your guests. Our catering menu includes:	What kind of catering services does The Daily Grind offer, and how far in advance should customers place catering orders?	The Daily Grind offers catering services for both coffee and baked goods, perfect for corporate meetings, family gatherings, or special celebrations. Our menu includes coffee urns (hot or cold), assorted pastry platters, custom cake orders, and breakfast and lunch sandwich platters (pre-order	Success

	* Coffee urns (hot or cold) with all necessary accompaniments. * Assorted pastry platters. * Custom cake orders. * Breakfast and lunch sandwich platters (pre-order required). Please contact us at least 48 hours in advance for small orders and 1 week for larger events to ensure availability and proper preparation. A catering specialist will be happy to assist you with your order.		required). For small orders, please contact us at least 48 hours in advance, and for larger events, a week's notice is recommended to ensure availability and proper preparation. A catering specialist will be happy to assist you with your order.	
8.	# Employment Opportunities The Daily Grind is always looking for passionate and dedicated individuals to join our team. We offer a supportive work environment, opportunities for growth, and competitive compensation. We regularly hire for positions such as:	What types of employment opportunities are typically available at The Daily Grind, and how can interested individuals apply?	The Daily Grind offers several employment opportunities, including: - *Barista*: Experience is preferred but not required, as training is provided. - *Baker*: Professional kitchen experience is required. - *Customer Service	Success

	* Barista: Experience preferred but not required; training provided. * Baker: Experience in a professional kitchen setting required. * Customer Service Associate: Friendly and enthusiastic individuals welcome. If you are interested in a career with us, please submit your resume and a cover letter to careers@thedailygrind.com. We look forward to hearing from you!		Associate*: Friendly and enthusiastic individuals are welcome. If you're interested in joining the team, you can submit your resume and a cover letter to careers@thedailygrind.com. We look forward to hearing from you!	
9.	# In-Store Experience We strive to make your visit to The Daily Grind as comfortable and enjoyable as possible. Our store features: * Free Wi-Fi: Perfect for working or Browse. * Comfortable Seating: A mix of tables, chairs, and cozy armchairs.	What amenities and features are available for customers to enjoy while visiting The Daily Grind?	At The Daily Grind, we offer several amenities to enhance your visit. You can enjoy free Wi-Fi, perfect for working or browsing. We provide comfortable seating with a mix of tables, chairs, and cozy armchairs. During the right season, you can relax on our outdoor patio. We also have	Success

	* Outdoor Patio: Seasonal seating available for enjoying the weather. * Restrooms: Clean and accessible facilities for customers. * Music: A curated playlist to enhance the relaxed atmosphere. We also have a small selection of local newspapers and magazines available for our customers' enjoyment.		clean and accessible restrooms for our customers. To add to the ambiance, we play a curated playlist to create a relaxed atmosphere. Additionally, we have a small selection of local newspapers and magazines available for your enjoyment.	
10.	# Community Engagement The Daily Grind is more than just a coffee shop; we are an active member of the Anytown community. We believe in giving back and fostering local connections. Our community initiatives include: * Sponsoring local school events and sports teams. * Hosting regular "Open Mic Nights" for local artists.	How does The Daily Grind engage with and support the Anytown community?	The Daily Grind is deeply committed to engaging with and supporting the Anytown community in several ways. We sponsor local school events and sports teams, host "Open Mic Nights" for local artists, and display artwork from local artists on a rotating basis. We also partner with local charities for fundraising events and use eco-friendly practices while	Success

	* Displaying artwork from local artists on a rotating basis. * Partnering with local charities for fundraising events. * Using eco-friendly practices and supporting local suppliers. We are always open to new ideas for community involvement. Please reach out if you have a proposal!		supporting local suppliers. Our goal is to foster local connections and give back to the community that supports us. If you have any ideas for community involvement, we're always open to hearing them!	
11.	# Payment Methods For your convenience, The Daily Grind accepts a variety of payment methods. We aim to make your checkout process smooth and efficient. We accept: * Cash * All major credit cards: Visa, Mastercard, American Express, Discover * Debit cards * Mobile payments: Apple Pay, Google Pay * The Daily Grind Gift Cards	What payment methods are accepted at The Daily Grind?	At The Daily Grind, we accept a variety of payment methods to make your checkout process smooth and convenient. You can pay with cash, all major credit cards (Visa, Mastercard, American Express, Discover), debit cards, mobile payments (Apple Pay, Google Pay), and The Daily Grind Gift Cards. Please note that we do not accept personal checks.	Success

	Please note that we do not accept personal checks.			
12.	# Health and Safety Protocols The health and safety of our customers and staff are our top priority at The Daily Grind. We adhere to all local health guidelines and maintain strict hygiene practices. Our protocols include: * Regular cleaning and sanitization of all surfaces. * Staff training on food safety and hygiene. * Gloves and masks worn by food handlers. * Availability of hand sanitizer for customers. * Clear signage regarding any current health advisories. We are committed to providing a clean and safe environment for everyone. Please do not hesitate to speak with a staff member if you have any concerns.	What health and safety protocols does The Daily Grind have in place to ensure the well-being of its customers and staff?	At The Daily Grind, we prioritize the health and safety of our customers and staff. Our protocols include regular cleaning and sanitization of all surfaces, staff training on food safety and hygiene, and the use of gloves and masks by food handlers. We also provide hand sanitizer for customers and display clear signage regarding any current health advisories. We are committed to maintaining a clean and safe environment for everyone. If you have any concerns, please feel free to speak with a staff member.	Success

13.	None - Should answer that the answer to the question is not available	"What is the average daily foot traffic at The Daily Grind?"	I don't have the exact number for the average daily foot traffic at The Daily Grind, but we do get a steady stream of customers throughout the day. Our busiest times are usually in the morning and around lunchtime. If you have any specific questions or need more detailed information, feel free to ask!	Success
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The recording of all the testing instances above is accessible at the Google Drive folder:

https://drive.google.com/drive/folders/1h64wvIdPOWSqImVpyEDQVm_8x5sGgt2z?usp=drive_link

Effectiveness of Marketing Campaigns: AI-generated responses under 5 seconds.

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Generate AI-suggested offer message	Generate the message	(14.45.10) The seller clicks on the generate AI offer button (14.45.15)	The AI message would appear onto the fields	The application instantly populates the text fields with	01/06/2025	Success (only about 3-4 seconds)

			Message populated		AI-generated message		
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System Uptime and Reliability: running without major crashes during a 1-hour continuous testing period.

Notes	Topic	Scenario	Test Steps	Expected Result	System Response	Date Tested	Status
Positive Testing	Run Test	Testing all features	(15.00.00) The seller performs auth (15.10.00) The seller builds & manages the agent, chunks (15.20.00) The seller tries the RAG bot (15.30.00) The seller feeds 50+ invoices (15.50.00) The seller conducts clustering (16.00.00)	The full-flow of app ran without critical failure	The application supports full-flow cycle of the app	01/06/2025	Success

			The seller conducts campaign				
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2. Qualitative testing

a. User Acceptance Testing

User Acceptance Testing (UAT) is a phase of software development where a trial of the software product is done by the intended audience (Gillis, 2022). UAT's goal is to ensure that the software can handle the targeted tasks in the real world and perform up to the user's requirements.

In the scenario of Selly, the UAT is done face-to-face during a visit to GoTo's office located at Pasaraya, Jakarta. There are several useful pieces of feedback that we received and implemented from the stakeholders.

The results of the UAT that were acknowledged by the Product Development Staff at GoTo signify that all of the features of this product have worked successfully. On top of that, the results also state that all the problems included in the problem statement have been solved, which is the lack of intelligent, AI-driven tools needed to stay relevant in a competitive market, as well as limited automation and static features that fail to address the dynamic demands of modern sellers.

b. Proof of User Acceptance Test

User Acceptance Test (Face-to-face testing at GoTo Head Office, Pasaraya)	 President University Approval
Project name: Driving SME Growth with AI-Based Seller Support Solutions for Selly	

User Acceptance Test (Feature)				
No	Use Cases / Processes	Acknowledged by	Test date	Status
1.	Login and register	Mr. Reza	17/06/25	Success
	Parameters :			
	Seller can register with a new email address			
	Seller can login with a registered email address and password			
2.	Add, edit, delete, view agent list, view agent detail, activate agent	Mr. Reza	17/06/25	Success
	Parameters :			
	Seller can add new agent			
	Seller can edit agent			
	Seller can delete agent			
	Seller can view agents in a list			
	Seller can view agent details			
	Seller can activate agent			
3.	Add, and delete knowledge bases, view knowledge base list, view knowledge base detail, activate knowledge base	Mr. Reza	17/06/25	Success
	Parameters :			
	Seller can add a new knowledge base			

	Seller can edit knowledge base			
	Seller can delete knowledge base			
	Seller can view knowledge bases in a list			
	Seller can view knowledge base details			
	Seller can activate knowledge base			
4.	Add, edit, and delete chunks, view chunks in a list, view chunks in detail,	Mr. Reza	17/06/25	Success
	Parameters :			
	Seller can add a new chunk			
	Seller can edit a chunk			
	Seller can delete a chunk			
	Seller can view chunks in a list			
	Seller can view chunk details			
5.	Activate, send queries, and receive results on the keyboard interface,	Mr. Reza	17/06/25	Success
	Parameters :			
	Seller can activate the keyboard interface			
	Seller can send queries to the keyboard interface			
	Seller can receive results on the keyboard interface			
6.	Add, edit, and delete invoices, view invoices in a list, view invoice detail,	Mr. Reza	17/06/25	Success
	Parameters:			
	Seller can add a new invoice			
	Seller can edit invoice			

	Seller can delete invoice			
	Seller can view invoices in a list			
	Seller can view invoice details			
7.	Clusterize, view result, view list	Mr. Reza	17/06/25	Success
	Parameters:			
	Seller can conduct the clustering action			
	Seller can view cluster result			
	Seller can view clustered customers name list			
8.	Campaign, generate & send message	Mr. Reza	17/06/25	Success
	Parameters:			
	Seller can fill in their own promotional message			
	Seller can use AI-message generator			
	Seller can choose the target recipient			
	Seller can send the message to recipient via WhatsApp			

Key Points	Quotes and Analysis
Authentication	P1 commented, " <i>The process was simple, email and password worked instantly after registration.</i> " P2 added, " <i>This flow makes it accessible even to sellers unfamiliar with tech platforms.</i> " Clear validation confirms the login/register system is user-friendly for SMEs.
Agent Management (RAG Module)	P1 recalled an application that was themed in law that was used to replace Paralegals in quoting the exact information section. P1 Quoted " <i>.. it would be great if it could answer more concisely</i> ". It was immediately discussed that the desired effect can be achieved through commanding the agent. P2 later replied, " <i>It is very interesting indeed that the seller is given that much customizability from the system</i> ". The statement shows that sellers may not be aware of the fact that answer

	complexity can be configured through the agent command. A toggle for word count is later implemented to raise awareness.
Knowledge Base Management (RAG Module)	P2 said, <i>"It is good to have full control over which information goes to which chunk through the markdown header scheme."</i> P1 added, <i>"Seeing the detail view helps ensure data is right before activation."</i> The chunk visibility and editability support content accuracy.
Keyboard Interface Interaction	P1 commented, <i>"Copying a customer's question and getting an answer in seconds shows how useful this is."</i> P2 followed up, <i>"Feels like chatting with a smart assistant, quick and practical."</i> The interface proved intuitive and fast for business inquiries.
Multi-Language Support (RAG Module)	P1 realized an interesting point: <i>"Would the application work if the question is stated in a different language than the business information?"</i> . The feature is then tested immediately by stating a question in Indonesian, even though the business information was in English. The result showed a clear and accurate answer in Indonesia, depicting the capability of the system in accommodating different languages. At the end of the showcase, P2 once more stated: <i>"What an interesting system."</i>
Invoice Management (Campaign Module)	P2 simply responded with, <i>"Okay, this is already as expected,"</i> when the feature was presented. This indicates that the feature meets the basic requirements and functions as expected without any immediate issues or standout elements noted by the participant.
Customer Clustering (Campaign Module)	Participants were interested in how it managed to clusterize customers based on Recency, Frequency, and Monetary (RFM) values. P2 commented, <i>"It's impressive how it uses RFM to cluster them."</i> P2 provided constructive feedback, adding, <i>"It would be great if there was a toggle slider to re-edit the value of the RFM classes for loyal, moderate, and at-risk customers."</i>
Campaign Generator (Campaign Module)	The campaign feature was well-received. P1 stated, <i>"This is innovative!"</i> However, P1 also added a suggestion: <i>"It would be great if we can set the end-goal because in every business we have our own end-goal, not just for the sake of a sale to happen."</i>

Jakarta, June 30, 2025
Acknowledged by,

A handwritten signature in black ink, appearing to be 'Reza', on a light gray rectangular background.

Reza Ghazali

3. Detail the test procedures that are carried out according to the design.

(Already performed in each table testing process)

4. Procedures for the demo are created and verified.

(Already performed in each table testing process)

D. TESTING OTHER SPECIFICATIONS:

Security, must hash user password

email varchar	encrypted_password varchar
newaccount@gmail.com	\$2a\$10\$CFw4u7tTeQJ64XTRKZvOS.8OZ
jojaya@gmail.com	\$2a\$10\$1a1yLxTAf2SvZYUwsD9Wweetb8
newaccount2@gmail.com	\$2a\$10\$IfGLolsF4/C6EFyCE2J9PONvwX
akunbarulagi@gmail.com	\$2a\$10\$Ap/ThpSqJ8qi0EejAKSz7ePvyw
newuser@selly.com	\$2a\$10\$XpbmQu85C9uSsen/epOoGOvN
josephthedja.nugrahawibawa@mattel.com	\$2a\$10\$UgtjUOL8DGum62DfHcBLae22B
jayaiskandar17@gmail.com	\$2a\$10\$XF2T7wgJa/l.rT7fB5cADeWMEh
jayattyl@selly.com	\$2a\$10\$LBPJDmt1DNEhkPP/BKLLJ.X1l3E
broski@gmail.com	\$2a\$10\$QQE/j8GkLGIPrcJjNWYwxedar
jayaiskandar21@gmail.com	\$2a\$10\$kK5k2WKWLd96z4xKyV73weBhl
jojaya@gmail.com	\$2a\$10\$aOAVj2SggvrVG/.kRARYej4eTir
usernew@selly.com	\$2a\$10\$pH1V/AT74pEKuJj45AR9P.U96G5
josephthedjanw@gmail.com	\$2a\$10\$sWEP6dV3ahievgpLQYAnm.DSa.
newaccount3@gmail.com	\$2a\$10\$RO2JFb3Kukrj4BqUNB0hMem1o
akunbaru@gmail.com	\$2a\$10\$C1Fyj.MyEToB6RZyvoFbXef0NIU

Record of the creation of the account newaccount3@gmail.com is recorded and accessible in the google drive link below:

https://drive.google.com/file/d/1DbX29_6z4awcGgTHew4skU3E0jjHs-QY/view?usp=drive_link

Data transaction, logged response time

Jun 28, 2025, 08:34pm				
28 Jun 21:33:47	200	POST	/rest/v1/rpc/fetch_agents	
28 Jun 21:33:44	200	GET	/rest/v1/invoices	
28 Jun 21:33:44	200	POST	/auth/v1/token	
28 Jun 21:33:43	200	POST	/auth/v1/token	
28 Jun 21:29:00	200	GET	/rest/v1/invoices	
28 Jun 21:28:58	200	GET	/rest/v1/invoices	
28 Jun 21:28:58	200	GET	/auth/v1/user	
28 Jun 21:28:56	200	GET	/rest/v1/invoices	
28 Jun 21:24:51	200	GET	/rest/v1/invoices	
28 Jun 21:24:48	200	GET	/rest/v1/invoices	
28 Jun 21:24:48	200	GET	/auth/v1/user	
28 Jun 21:24:45	200	POST	/rest/v1/customer_segments	
28 Jun 21:24:45	200	GET	/auth/v1/user	

Jun 28, 2025, 08:34pm				
28 Jun 21:31:45	200	POST	9b422208-da1f-4258-88ed-8c97965605ed	
28 Jun 21:29:09	200	POST	d5271564-669e-4be8-907b-8424c8f65ef8	

E. CONCLUSION

Based on the testing results, it is concluded that it has passed all of the required tests which were also defined priorly in the product requirement document in terms of how it should answer towards the stakeholder's pain point. The product has been validated one-on-one too with the stakeholder for assessing the values delivered by this application, which were presented through the Auto Text RAG and Personalized Marketing Campaign Generator feature. This successful validation confirms the product is not only fit for purpose but also strategically aligned with stakeholder needs.

F. REFERENCE

Gillis, A. (2022, March 14). *What is User Acceptance Testing (UAT)? | Definition from TechTarget*. Search Software Quality.
<https://www.techtarget.com/searchsoftwarequality/definition/user-acceptance-testing-UAT>